Project: Test a Perceptual Phenomenon

Background Information:

In a Stroop task, participants are presented with a list of words, with each word displayed in a color of ink. The participant's task is to say out loud the *color of the ink* in which the word is printed. The task has two conditions: a congruent words condition, and an incongruent words condition. In the *congruent words* condition, the words being displayed are color words whose names match the colors in which they are printed: for example RED, BLUE. In the *incongruent words* condition, the words displayed are color words whose names do not match the colors in which they are printed: for example PURPLE, ORANGE. In each case, we measure the time it takes to name the ink colors in equally-sized lists. Each participant will go through and record a time from each condition.

Dataset:

The dataset contains results from a number of participants in the task. Each row of the dataset contains the performance for one participant; with the first number their results on the congruent task and the second number their performance on the incongruent task.

Congruent	Incongruent
12.079	19.278
16.791	18.741
9.564	21.214
8.63	15.687
14.669	22.803
12.238	20.878
14.692	24.572
8.987	17.394
9.401	20.762
14.48	26.282
22.328	24.524
15.298	18.644
15.073	17.51
16.929	20.33
18.2	35.255
12.13	22.158
18.495	25.139
10.639	20.429
11.344	17.425
12.369	34.288
12.944	23.894
14.233	17.96

19.71	22.058
16.004	21.157

Investigation:

1. What is our independent variable? What is our dependent variable?

Independent variable: Two test conditions: a congruent words condition, and an incongruent words condition

Dependent variable: Time taken to name the ink colors from each condition.

2. What is an appropriate set of hypotheses for this task? What kind of statistical test do you expect to perform? Justify your choices.

Hypotheses: We want to see if there is a significant difference between the times taken in Congruent & Incongruent test. Our NULL hypothesis is there is no significant difference. The mean time taken by the population for congruent test is similar to incongruent test. Alternative hypothesis is that they are not equal.

$$H_0: \mu_{Congruent} = \mu_{Incongruent}$$

$$H_A: \mu_{Congruent} \neq \mu_{Incongruent}$$

Statistical test we should perform: Dependent t-test for paired samples (two conditions)

Justification of performing t-test rather than z-test: z-test works when we know population parameters μ and σ. Here we don't have population parameters; we have only samples to draw our conclusions. So we will estimate population standard deviation using sample standard deviation. Hence we get a sampling distribution which is more prone to error and have ticker tail (than normal distribution) and it is called t-distribution. We will perform t-test to calculate t-score and derive p-value of the sample (from t-table).

T-test assumes population data to be approximately normally distributed. It is known that non normal populations possessing nearly bell-shaped distributions provides values of T that approximate the t-distribution very closely. From the histogram drawn below we can see that samples from congruent test and incongruent test forms a nearly bell-shaped curve. So the sampling distribution of individual differences of congruent & incongruent time also approximately forms bell-shaped distribution that approximate the t-distribution very closely.

Justification of performing dependent t-test: Here 2 samples are dependent as the same subject (person) takes the test twice in two conditions.

3. Report some descriptive statistics regarding this dataset. Include at least one measure of central tendency and at least one measure of variability.

Descriptive Statistics of Congruent Test:

Average time taken for congruent test $\overline{X}_{Congruent} = 14.05$ with Standard Deviation $S_{Congruent} = 3.56$

Descriptive Statistics of Incongruent Test:

Average time taken for congruent test $\overline{X}_{Incongruent} = 22.01$ with Standard Deviation $S_{Incongruent} = 4.79$

4. Provide one or two visualizations that show the distribution of the sample data. Write one or two sentences noting what you observe about the plot or plots.

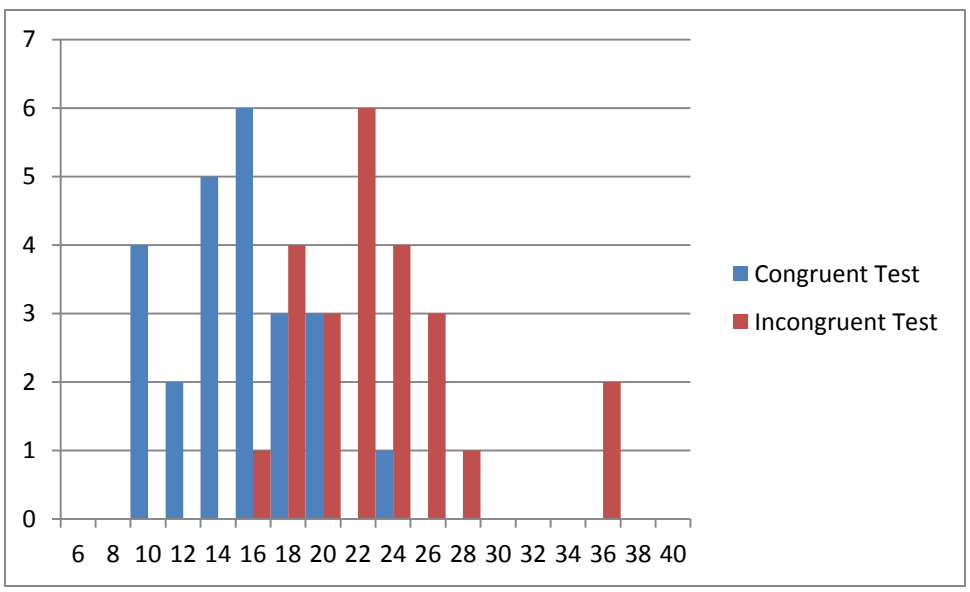


Figure: Histogram of Congruent & Incongruent test results.

Observations:

1. Congruent test has average time 14.05 & Mode is the bucket (14 to 16)

2. Incongruent test has average 22.01 & Mode is the bucket (20 to 22)
3. There are no outliers.

5. Now, perform the statistical test and report your results. What is your confidence level and your critical statistic value? Do you reject the null hypothesis or fail to reject it? Come to a conclusion in terms of the experiment task. Did the results match up with your expectations?

T-Statistics:

1. Difference of mean between the populations of Congruent & Incongruent tests: $M_D = \overline{X}_{Congruent} - \overline{X}_{Incongruent} = -7.9648$

2. Standard deviation of the time difference for each person in congruent & incongruent test $S_D = S_{Congruent-Incongruent} = 4.8648$

3. Calculate t-statistics, n=24 (df = 23): $t = \frac{\overline{X}_{Congruent} - \overline{X}_{Incongruent}}{S_D/\sqrt{n}} = -8.02$

4. For n=24 (df = 23), t-Critical values for α=0.05 for two tailed test: $t = \pm 2.069$

Decision:

1. t_statistics = -8.02, t_critical = ± 2.069
2. We reject H_0.
3. So a person took significantly more time in incongruent test compared to congruent test.

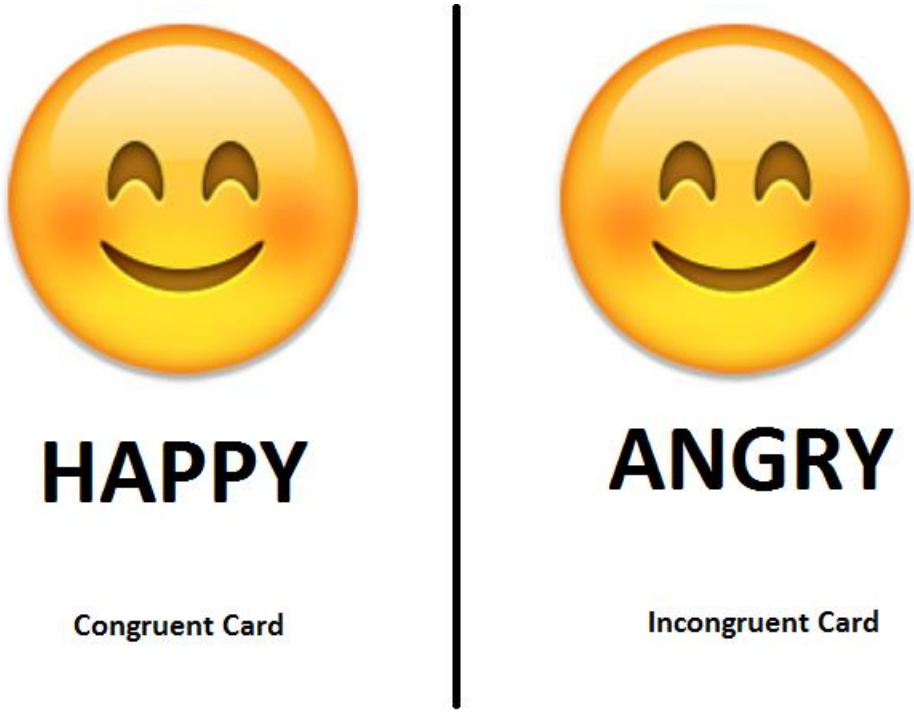
Effect size measure using Cohen's d:

Cohen's d = $\frac{M_D}{S_D} = \frac{-7.9648}{4.8684} = -1.6372$

95% Confidence Interval for mean population difference:

$$MD \pm t_{critical} * \left({S_D}/{\sqrt{n}} \right) = (-5.88, -10.05)$$

6. Optional: What do you think is responsible for the effects observed? Can you think of an alternative or similar task that would result in a similar effect? Some research about the problem will be helpful for thinking about these two questions!

In case of incongruent condition, the interference between the different information our brain receives (what the words say and the color of the words) causes more time to give answer.

We can design similar experiment with emoji & emotion written in a card. For congruent case emotion corresponds to the emoji, but incongruent case emotion doesn't correspond to the emoji. **A person needs to tell the emotion written in the card.**